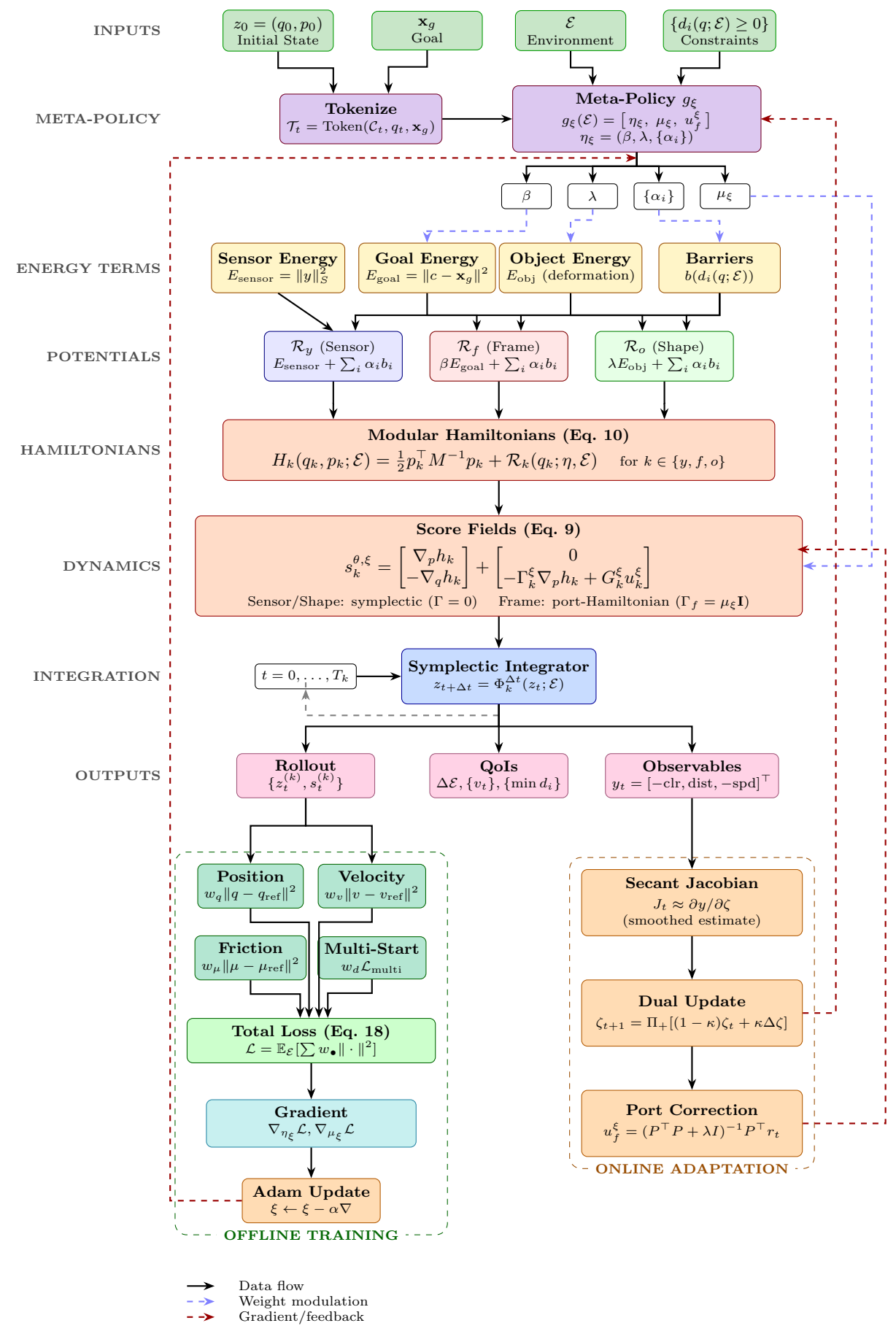
INPUTS
$z_0 = (q_0, p_0)$ Initial State
$\mathbf{x}_g$ Goal
$\mathcal{E}$ Environment
$\{d_i(q;\mathcal{E}) \geq 0\}$ Constraints
META-POLICY
Tokenize
$\mathcal{T}_t = \text{Token}(\mathcal{C}_t, q_t, \mathbf{x}_g)$
Meta-Policy g_ξ
$g_\xi(\mathcal{E}) = [\eta_\xi,\ \mu_\xi,\ u_f^\xi]$
$\eta_\xi = (\beta, \lambda, \{\alpha_i\})$
β
λ
$\{\alpha_i\}$
μ_ξ
ENERGY TERMS
Sensor Energy
$E_{\text{sensor}} = \|y\|_S^2$
Goal Energy
$E_{\text{goal}} = \|c - \mathbf{x}_g\|^2$
Object Energy
E_{obj} (deformation)
Barriers
$b(d_i(q;\mathcal{E}))$
POTENTIALS
$\mathcal{R}_y$ (Sensor)
$E_{\text{sensor}} + \sum_i \alpha_i b_i$
$\mathcal{R}_f$ (Frame)
$\beta E_{\text{goal}} + \sum_i \alpha_i b_i$
$\mathcal{R}_o$ (Shape)
$\lambda E_{\text{obj}} + \sum_i \alpha_i b_i$
HAMILTONIANS
Modular Hamiltonians (Eq. 10)
$H_k(q_k, p_k; \mathcal{E}) = \frac{1}{2} p_k^\top M^{-1} p_k + \mathcal{R}_k(q_k; \eta, \mathcal{E})$ for $k \in \{y, f, o\}$
DYNAMICS
Score Fields (Eq. 9)
$s_k^{\theta,\xi} = \begin{bmatrix} \nabla_p h_k \\ -\nabla_q h_k \end{bmatrix} + \begin{bmatrix} 0 \\ -\Gamma_k^\xi \nabla_p h_k + G_k^\xi u_k^\xi \end{bmatrix}$
Sensor/Shape: symplectic ($\Gamma = 0$) Frame: port-Hamiltonian ($\Gamma_f = \mu_\xi \mathbf{I}$)
INTEGRATION
$t = 0, \ldots, T_k$
Symplectic Integrator
$z_{t+\Delta t} = \Phi_k^{\Delta t}(z_t; \mathcal{E})$
OUTPUTS
Rollout
$\{z_t^{(k)}, s_t^{(k)}\}$
QoIs
$\Delta\mathcal{E}, \{v_t\}, \{\min d_i\}$
Observables
$y_t = [-\text{clr}, \text{dist}, -\text{spd}]^\top$
Position
$w_q \|q - q_{\text{ref}}\|^2$
Velocity
$w_v \|v - v_{\text{ref}}\|^2$
Friction
$w_\mu \|\mu - \mu_{\text{ref}}\|^2$
Multi-Start
$w_d \mathcal{L}_{\text{multi}}$
Total Loss (Eq. 18)
$\mathcal{L} = \mathbb{E}_\mathcal{E}[\sum w_\bullet \| \cdot \|^2]$
Gradient
$\nabla_{\eta_\xi} \mathcal{L}, \nabla_{\mu_\xi} \mathcal{L}$
Adam Update
$\xi \leftarrow \xi - \alpha \nabla$
OFFLINE TRAINING
Secant Jacobian
$J_t \approx \partial y / \partial \zeta$
(smoothed estimate)
Dual Update
$\zeta_{t+1} = \Pi_+[(1-\kappa)\zeta_t + \kappa \Delta\zeta]$
Port Correction
$u_f^\xi = (P^\top P + \lambda I)^{-1} P^\top r_t$
ONLINE ADAPTATION
Data flow
Weight modulation
Gradient/feedback